$2C + 2D = 100$

$C + D = 50$ .....(iii)

And also $C = \frac{D}{2}$

$\frac{D}{2} + D = 50$

$3D = 100$

$D = \frac{100}{3}$

$C = \frac{100}{3 \times 2} = \frac{100}{6} = \frac{50}{3}$

$A = D + 5$

$= \frac{100}{3} + 5 = \frac{115}{3}$

Now

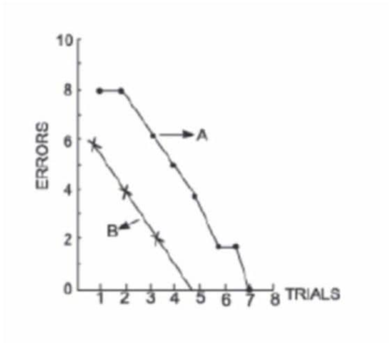
$B = 100 - A - C - D$

$= 100 - \frac{115}{3} - \frac{50}{3} - \frac{100}{3}$

$B = \frac{35}{3}$

It is very clear that Alok has the most money.

133. Given below is the performance of two learners A and B:



Which one of the following can be deduced from the graph?

(a) The learning task was difficult

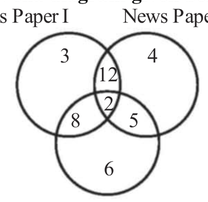
(b) B learned faster than A

(c) B learned faster than A throughout the task

(d) A learned faster than B throughout the task

Ans. (c) : It is very clear from the graph that B commits less error as compared to A throughout the task. So B learned faster than A throughout the task.

134. Consider the Vena diagram given below:



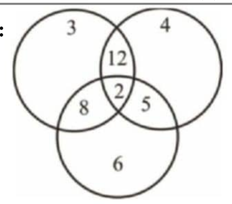
The numbers In the Venn diagram are persons reading the newspapers indicated. The diagram is drawn after surveying 50 persons. In a population of 10,000, how many can be expected to read at least two newspapers?

Choose the correct answer from the codes given below:

(a) 5000 (b) 6250

(c) 6000 (d) 5400

Ans. (d) :



50 persons reading atleast two newspapers.

$= 12 + 2 + 8 + 5 = 27$

1 person $= \frac{27}{50}$

So for 10,000 person $= \frac{27}{50} \times 10{,}000 = 5400$

135. In an examination every candidates took either physics or Mathematics or both. 65.8% of the candidates took physics and those who took Mathematics were 59.2% . The total candidates were 2000. How many candidates took both mathematics and physics?

(a) 750 (b) 700

(c) 550 (d) 500

Ans. (d) : Total numbers of candidates are = 2000

% of candidates who take physics = 68. 5%

No. of candidate $= \frac{68.5}{100} \times 2000$

Who took physics = 1316

% of person who took mathematics out of 2000 are

$= \frac{59.2}{100} \times 2000$

Total number of candidate, who took physics and mathematics are = 1316 + 1184 = 2500

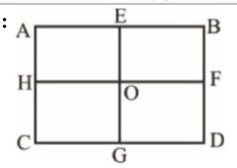
∴ The number of candidates who took both mathematics and physics = 2500 – 2000 = 500

136. A point to selected at random inside a rectangle and perpendiculars are drawn on each side from the point. The sum of these perpendicular is 24 cm. If the length of rectangle is 3 times the width, then what will be the perimeter of the rectangle?

(a) 36 cm (b) 42 cm

(c) 24 cm (d) 48 cm

Ans. (d) :



Rectangle ABCD has width = x

Length = $3x$

But, also

From point O, perpendicular are drawn on each side and there sum in 24 cm.

Now,

$\frac{x}{2} + \frac{x}{2} + \frac{3x}{2} + \frac{3x}{2} = 24$

$x + 3x = 24$

$x = 6$ cm

Length = $6 \times 3 = 18$ cm

Width = 6

Perimeter of Rectangle = $2(l + b)$

$= 2[18 + 6]$

$= 2[24]$

Perimeter = 48 cm

137. If the product of two numbers is 320 and their ratio is 1 : 5, what is the difference between the squares of these two numbers ?

(a) 1024 (b) 1365

(c) 1536 (d) 1635

Ans. (c) : Let the two number be x and y

$x \times y = 320$ .....(i)

and also

$\frac{x}{y} = \frac{1}{5}$

$5x = y$ .....(ii)

Putting equ^{n}. (ii) to equ^{n}. (i)

$x \times 5x = 320$

$5x^2 = 320$

$x^2 = 64$

$y = 8 \times 5$

The difference between square of these number is

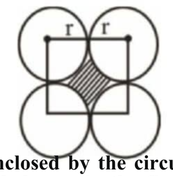
$y^2 - x^2$

$= (40)^2 - (8)^2$

$= 1600 - 64$

$= 1536$

138. Four equal circles are described at the four corners of a square so that each touches two of the others as shown in the figure:

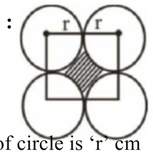


The area enclosed by the circumferences of the circles (shaded area in the figure) is 13(5/7) sq. cm. Find the radius of the circles. (Assume = p = 22/7)

(a) 4 cm (b) 2.5 cm

(c) 7.5 cm (d) 6 cm

Ans. (a) :



Radius of circle is 'r' cm

Area of shaded region $= 13\frac{5}{7} \text{cm}^2$

As per given figure

Area of shaded region = Area of square 4 × quarter of circle area

$= (2r)^2 - 4 \times \frac{\pi r^2}{4} = 4r^2 - \pi r^2$

$\frac{96}{7} = r^2 \left[4 - \frac{22}{7}\right]$

$r^2 = \frac{96}{7} \times \frac{7}{6}$

$r^2 = 16$

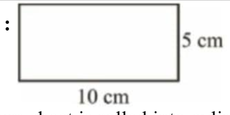
$r = 4$

139. A rectangular sheet of aluminum foil measuring 5 cm by 10 cm is rolled into a cylinder of height 10 cm. If the cost of painting the cylinder is Rs. 2/per sq cm, then the cost of painting the surface of the cylinder will be:

(a) Rs. 400 (b) Rs. 200

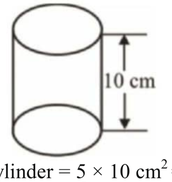
(c) Rs. 150 (d) Rs. 100

Ans. (d) :



Aluminium sheet is rolled into cylinder of height 10 cm.

So



Surface Area of cylinder = $5 \times 10 \text{ cm}^2 = 50 \text{ cm}^2$

Cost of painting is 2 Rs./cm^2, so $50 \text{ cm}^2 \times 2 \text{ rs/cm}^2$

= Rs. 100